

David R. Miller

PHD CANDIDATE - UNIVERSITY OF BRITISH COLUMBIA

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Education

University of British Columbia

Vancouver, BC, Canada

PHD, ASTRONOMY

2022–Present

Supervisors: Dr. Jeremy Heyl and Dr. Harvey Richer (1944–2023)

Thesis: Probing Stellar Evolution With White Dwarfs in Open Clusters

Expected Completion: Spring 2027

University of British Columbia

Vancouver, BC, Canada

MSc, ASTRONOMY

2020–2022

Supervisor: Dr. Harvey Richer (1944–2023)

Thesis: The White Dwarf Initial-Final Mass Relation: Ultramassive White Dwarfs Ejected from Young Open Star Clusters

Simon Fraser University

Burnaby, BC, Canada

BSc, PHYSICS HONOURS (WITH DISTINCTION)

2016–2020

Supervisor: Dr. Levon Pogorian

Thesis: Dark Sector Cosmic Fluid Interactions as a Solution to the Observed Disagreement in Measurements of the Hubble Expansion Rate

Awards

UBC Four Year Fellowship (4YF), \$98,000

2023–2027

President's Academic Excellence Initiative PhD Award, \$4,800

2022–2026

Dunlap Institute Summer Undergraduate Research Program Fellowship (SURP), \$9,500

2019

NSERC Undergraduate Student Research Award (USRA), \$5,250

2018

Ken Caple College Transfer Entrance Scholarship, \$3,500

2016

Publications

First-Author Publications (4)

1. **Miller, D. R.**, Heyl, J., and Tremblay, P-E. Investigating the Observational Progenitor Mass Gap in the White Dwarf Initial-Final Mass Relation. I. Cluster Census and Characterization of the First White Dwarfs in the Gap. 2026, *ApJ*, accepted.
2. **Miller, D. R.**, Caiazzo, I., Heyl, J., et al. The White Dwarf Initial-Final Mass Relation from Open Clusters in Gaia DR3. 2026, *ApJ*, 996:69.
3. **Miller, D. R.**, Caiazzo, I., Heyl, J. et al. An Extremely Massive White Dwarf Escaped From the Hyades Star Cluster. 2023, *ApJL*, 956:L41.
4. **Miller, D. R.**, Caiazzo, I., Heyl, J. et al. The Ultramassive White Dwarfs of the Alpha Persei Cluster. 2022, *ApJL*, 926:L24.

Co-Author Publications (7)

1. Jadhav, V. V., Kroupa, P., **Miller, D. R.**, Sahu, S., and Franktisek, D. The white dwarf population of open

clusters and their tidal tails: Tracers of contamination and stellar interactions. 2026, *A&A*, accepted.

2. Yan, H., **Miller, D. R.**, Zhao, J., Guo, J., and Li, C. Escaped White Dwarf Candidates from Open Clusters. 2026, *ApJS*, 286:5.
3. Yan, H., Wang, L., **Miller, D. R.**, et al. Searching for White Dwarf Candidates Formed Through Binary Evolution in Star Clusters. 2026, *ApJ*, 999:167.
4. Cristea, A. A., Caiazzo, I., Cunningham, T., Raymond, J. C., Vennes, S., Kawka, A., Desai, A., **Miller, D. R.**, et al. A half ring of ionized circumstellar material trapped in the magnetosphere of a white dwarf merger remnant: A new class of white dwarf merger remnants with X-ray emission. 2026, *A&A*, 706, A188.
5. Côté, P., et al. (including **Miller, D. R.**). The CASTOR mission. 2025, *JATIS*, 11(4).
6. Richer, H. B., Caiazzo, I., Du, H., Grondin, S., Hegarty, J., Heyl, J., Kerr, R., **Miller, D. R.**, and Thiele, S. Massive White Dwarfs in Young Star Clusters. 2021, *ApJ*, 912:165. (Author order alphabetical after the first two)
7. Tzanavaris, P., Gallagher, S. C., Ali, S., **Miller, D. R.**, Pentering, S., and Johnson, K. E. Cosmic Pathways for Compact Groups in the Milli-Millennium Simulation. 2019, *ApJ*, 871:242.

Unpublished and In Preparation Work

1. **Miller, D. R.**, Yan, H., et al. Characterization of Candidate High-Mass White Dwarf Escapees Candidates from Open Clusters. In prep.
2. Yan, H., **Miller, D. R.**, Li, C., et al. Photometric and Spectroscopic Analysis of an Extremely Low-Mass White Dwarf Binary. In prep.
3. **Miller, D. R.**, Caiazzo, I., and Heyl, J. Atypical White Dwarfs in Open Clusters. In prep.
4. **Miller, D. R.**, Heyl, J. A Spectroscopic Census of Cluster White Dwarfs from DESI, LAMOST, and SDSS. In prep.
5. Heyl, J., Caiazzo, I., Richer, H., and **Miller, D. R.**. Reconstructing Nearby Young Clusters with Gaia EDR3. *arXiv pre-prints* arXiv:2110.04296.

Select Talks and Presentations

Klosterneuburg, Austria	24th European White Dwarf Workshop , ISTA	Contributed Talk	August 2026
Halifax, Canada	CASCA (Canadian Astronomical Society) 2025	Contributed Talk	June 2025
Burnaby, Canada	Cosmology Seminar , Simon Fraser University	Invited Talk	March 2025
Barcelona, Spain	23rd European White Dwarf Workshop	Contributed Talk	July 2024
Padova, Italy	EAS (European Astronomical Society) 2024	e-Poster	July 2024
Toronto, Canada	CASCA 2024: Harvey Richer Special Memorial Session	Invited Talk	June 2024
Loon Lake, Canada	Astronomy Retreat , University of British Columbia	Contributed Talk	March 2024
Vancouver, Canada	Astronomy Seminar , University of British Columbia	Invited Talk	Oct. 2022
Tübingen, Germany	22nd European White Dwarf Workshop	Contributed Talk	Aug. 2022

Observing Programs as Principle Investigator

Principle investigator on 15 successful observing programs

Gemini	<i>Spectroscopic confirmation of escaped high-mass white dwarfs for the cluster initial-final mass relation</i>	2026
Gemini	<i>Expanding the high-mass end of the cluster white dwarf initial-final mass relation</i>	2026
Co-PI: HST	<i>Hidden in Plain Sight? Unraveling the Potential Merger Histories of Ultra-Massive White Dwarfs in the Initial-Final Mass Relation</i>	2026
Gemini	<i>Tracing contamination and stellar interactions in the white dwarf populations of open clusters and their tidal tails</i>	2026
CFHT	<i>Probing the mass gap in the white dwarf initial-final mass relation with the rich intermediate-aged open cluster NGC 2360</i>	2026
Gemini	<i>Probing Accelerated Cooling and Formation in a White Dwarf Binary from a Young Open Cluster</i>	2026
CFHT	<i>Probing the white dwarf initial-final mass relation mass gap in intermediate age open clusters with CFHT MegaCam</i>	2025
Gemini	<i>Probing Non-Monotonic Trends in the Open Cluster White Dwarf Initial-Final Mass Relation</i>	2025
Gemini	<i>Developing the White Dwarf Initial-Final Mass Relation by Advancing the Survey of High-Mass White Dwarfs in Open Star Clusters</i>	2024
Gemini	<i>Improving the Census of High-Mass White Dwarfs in Open Star Clusters</i>	2024
Gemini	<i>High Mass White Dwarf Escapees From Clusters</i>	2023
Gemini	<i>An Extremely Massive White Dwarf Member of the vdB-Hagen 99 Cluster</i>	2022
Gemini	<i>An Ultramassive White Dwarf Escaped From the Hyades Cluster</i>	2021
Gemini	<i>Reconstructing Nearby Young Clusters with Gaia EDR3: Two White Dwarfs Escaped from Alpha Per</i>	2021
Gemini	<i>Spectroscopy of Massive White Dwarfs in Extended Survey Around Young Clusters</i>	2021

Teaching

Note: At the direction of my supervisor, I did not take on additional TA roles after 2023, as they were no longer expected following the award of a graduate fellowship. However, I was specifically requested to TA the graduate stellar astronomy course in 2026, a biennial offering, reflecting my role as the department's leading graduate student in stellar astrophysics.

Astro 508: Stellar Astronomy , TA	2026
Sole TA for this graduate course. Duties include holding office hours, supporting student learning, and managing grading.	
Astro 310: The Solar System , Head Tutorial TA	2022–2023
Oversaw tutorials, managed the course gradebook, held office hours, and revised tutorial materials now used in subsequent offerings.	
Astro 311: Stars and Galaxies , Tutorial TA	2023
Astro 102: Intro to Stars and Galaxies , Tutorial TA	2022
Phys 131: Energy and Waves , Tutorial TA	2021
Astro 101: Introduction to the Solar System , Tutorial TA	2020–2021
Ran and supported tutorials, held office hours, and marked coursework.	

Academic Service

Member

Canada

CASTOR SCIENCE TEAM - STELLAR ASTROPHYSICS WORKING GROUP

2022–present

- Contributed to the science case development for the Cosmological Advanced Survey Telescope for Optical and uv Research (CASTOR), which is a proposed Canadian-led UV-optical space telescope.

Member

University of British Columbia

PHYSICS & ASTRONOMY GRADUATE STUDENT ASSOCIATION

2020–present

- Participated in meetings and contributed feedback on matters affecting graduate students in the department.

Volunteer

Vancouver, BC, Canada

TESTING GRAVITY

2019, 2025

- Supported logistics for an international conference on testing the laws of gravity, including registration desk support, managing Zoom sessions, and equipment operation.

Sole Organizer

University of British Columbia

COMPACT OBJECT JOURNAL CLUB

2021–2024

- Organized and led weekly meetings, scheduled speakers, managed Zoom for remote attendees, and facilitated discussions.

Problem Setter and Judge

University of British Columbia

HIGH-ENERGY ASTROPHYSICS OLYMPIAD

2022

- Wrote questions for and judged a competitive physics Olympiad for upper division astronomy students.

Member

Simon Fraser University

PHYSICS UNDERGRADUATE STUDENT ASSOCIATION

2016–2020

- Attended regular meetings and provided input to enhance the undergraduate physics student experience.

Public Outreach

Speaker

Vancouver, BC, Canada

SPACE CAMP SATURDAYS – H. R. MACMILLAN SPACE CENTRE

2026

- Gave a public talk on stellar evolution for Space Camp Saturdays, an astronomy outreach program for families and young students.

Speaker

Vancouver, BC, Canada

SCIENCE RENDEZVOUS – UNIVERSITY OF BRITISH COLUMBIA

2026

- Delivered a public science talk on stellar evolution as part of Science Rendezvous, a Canada-wide free family science festival.

Volunteer

Vancouver, BC, Canada

ASTRONOMY ON TAP – UNIVERSITY OF BRITISH COLUMBIA

2024–present

- Helped coordinate the launch of the new UBC chapter of Astronomy on Tap, along with answering astronomy questions from participants.

Volunteer*Toronto, ON, Canada***ASTRONOMY ON TAP – UNIVERSITY OF TORONTO***2019*

- Supported event coordination and interacted with attendees to answer astronomy related questions.

Volunteer*University of Western Ontario***CRONYN OBSERVATORY***2018*

- Guided visitors and answered astronomy questions during public telescope viewing nights.

Professional Development

Participant*Virtual***SUMMER SCHOOL IN STATISTICS FOR ASTRONOMERS – PENN STATE CENTER***2026***FOR ASTROSTATISTICS AND ASTROINFORMATICS**

- Summer school on statistical methods for astronomical research, covering statistical inference, regression and model selection, resampling methods, Bayesian data analysis, MCMC, nested sampling, time series analysis, spatial statistics, and machine learning, with training conducted in R.

International Data Rights Holder*Canada***LEGACY SURVEY OF SPACE AND TIME (LSST) ON THE RUBIN OBSERVATORY***2025–present*

- Listed among those with provisional LSST data rights through Canada's in-kind contribution program, pending final data rights agreement.

Attendee*Vancouver, BC, Canada***ML4ASTRO WORKSHOP - ICML 2025***2025*

- Workshop on machine learning in astrophysics, covering foundation models, simulation-based inference, anomaly detection, and the integration of AI into scientific workflows.

Attendee*Vancouver, BC, Canada***TESTING GRAVITY PRE-CONFERENCE SCHOOL***2025*

- Review lectures and background material in gravitation and cosmology ahead of the Testing Gravity conference.

Participant*Montreal, QC, Canada***CRAQ SUMMER SCHOOL ON *Space Astronomy****2023*

- Covered Canadian capabilities in space-based astronomy, satellite operations, and mission and instrumentation development.

Attendee*Vancouver, BC, Canada***TESTING GRAVITY PRE-CONFERENCE SCHOOL***2019*

- Background lectures on cosmology and gravitation held prior to the Testing Gravity conference.

Participant*Montreal, QC, Canada***CRAQ SUMMER SCHOOL ON *Large-Scale Astrophysics****2018*

- Focused on galaxy formation and evolution, galaxy clusters, and the use of large-scale structures as cosmological tools.